MATH 152 **QUIZ 11 A** NAME Solutions

Mrs. Bonny Tighe 25 points

10.1,10.2,10.3 TA_____ Wed 12/6/2017

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 3\cos\theta,\ \ y = 4\sin\theta$

$\frac{x}{3} = \cos\theta,\ \ \frac{y}{4} = \sin\theta$

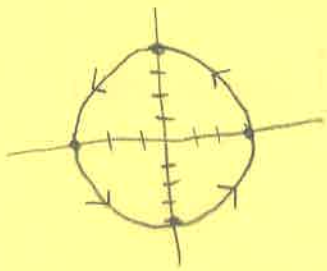
$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$

$\boxed{\frac{x^2}{9} + \frac{y^2}{16} = 1}$

To figure out direction:

θ	x	y
0	3	0
$\pi/2$	0	4

$(3,0) \to (0,4)$



b) $x = e^t - 3,\ \ y = e^{2t}$

$e^t = x+3$ $\quad = (e^t)^2$

$\boxed{y = (x+3)^2}$

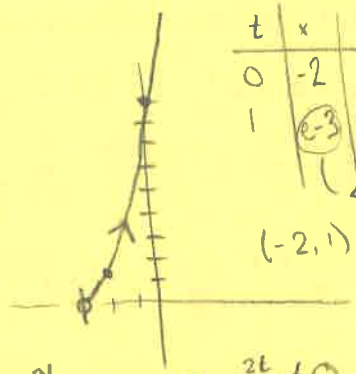
To figure out direction:

t	x	y
0	-2	1
1	$e-3$	e^2

>1

$<0, >-2$

$(-2,1) \to (e-3, e^2)$



Observe $y = e^{2t} \neq 0$ (hence the hole)

and $x = e^t - 3 > -3$ (hence no left half of parabola)

2. Find the equation of the tangent line to the curve $x = t^2 + 1,\ \ y = t^3$ at (2, - 1)

$x = t^2 + 1,\quad y = t^3$

$\frac{dx}{dt} = 2t \qquad \frac{dy}{dt} = 3t^2$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2}{2t} = \frac{3}{2}t$

$m = \left.\frac{dy}{dx}\right|_{t=-1} = \frac{3}{2}(-1) = -\frac{3}{2}$

What is t at $(x,y) = (2,-1)$?

$y = t^3$

$-1 = t^3 \Rightarrow t = -1$

Check: $x = (-1)^2 + 1 = 2$ ✓

$y - y_1 = m(x - x_1)$

$y + 1 = -\frac{3}{2}(x - 2)$

$\boxed{y = -\frac{3}{2}x + 2}$

3. Find the area enclosed by the x- axis and the curve $x = t^3 + 1,\ y = 3t - t^2$

$\hookrightarrow$ means $\int y\,dx$

Limits of integration:
When the curve crosses the x-axis

$y = 0 \Rightarrow 3t - t^2 = 0$
$t(3-t) = 0$
$t = 0, 3$

$x = t^3 + 1$
$dx = (3t^2)\,dt$

$$\int y\,dx = \int_0^3 (3t - t^2)(3t^2)\,dt$$
$$= \int_0^3 (9t^3 - 3t^4)\,dt$$
$$= \left[\tfrac{9}{4}t^4 - \tfrac{3}{5}t^5\right|_0^3$$
$$= \tfrac{9}{4}(3)^4 - \tfrac{3}{5}(3)^5 = 3^6\left(\tfrac{1}{4} - \tfrac{1}{5}\right) = \boxed{\frac{3^6}{20}}$$

4. Compute the surface area obtained by rotating the curve

$x = e^t + e^{-t},\ y = 3 + 2t\ ,\ 0 \le t \le 2$ about the y-axis. $S = 2\pi \int_\alpha^\beta radius(arc\ length)$

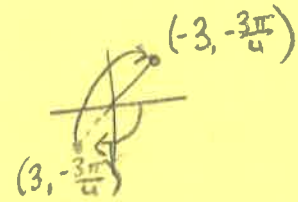
$r = x = e^t + e^{-t}$

$\frac{dx}{dt} = e^t - e^{-t},\ \frac{dy}{dt} = 2$

$$S = 2\pi \int_0^2 (e^t + e^{-t})\sqrt{(e^t - e^{-t})^2 + 4}\,dt$$
$$= 2\pi \int_0^2 (e^t + e^{-t})\sqrt{e^{2t} - 2 + e^{-2t} + 4}\,dt$$
$$= 2\pi \int_0^2 (e^t + e^{-t})\sqrt{(e^t + e^{-t})^2}\,dt$$
$$= 2\pi \int_0^2 (e^{2t} + 2 + e^{-2t})\,dt$$
$$= 2\pi \left(\tfrac{1}{2}e^{2t} + 2t - \tfrac{1}{2}e^{-2t}\right|_0^2$$
$$= 2\pi\left(\tfrac{1}{2}e^4 + 4 - \tfrac{1}{2}e^{-4} - \tfrac{1}{2} - 0 + \tfrac{1}{2}\right)$$
$$= \boxed{2\pi\left(\tfrac{1}{2}e^4 + 4 - \tfrac{1}{2}e^{-4}\right)}$$

"One of those"

5. a) Find the Cartesian coordinates for the polar coordinate $(-3, -3\pi/4)$



$x = r\cos\theta$
$= -3\cos(-\frac{3\pi}{4})$
$= -3(-\frac{\sqrt{2}}{2})$
$= \frac{3\sqrt{2}}{2}$

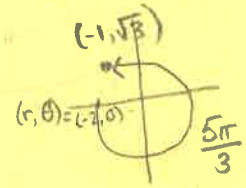
$y = r\sin\theta$
$= -3\sin(-\frac{3\pi}{4})$
$= -3(-\frac{\sqrt{2}}{2})$
$= \frac{3\sqrt{2}}{2}$

$\boxed{\left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)}$

b) Find polar coordinates for the Cartesian coordinate $(-1, \sqrt{3})$ where $r < 0, and\ \theta > 0$

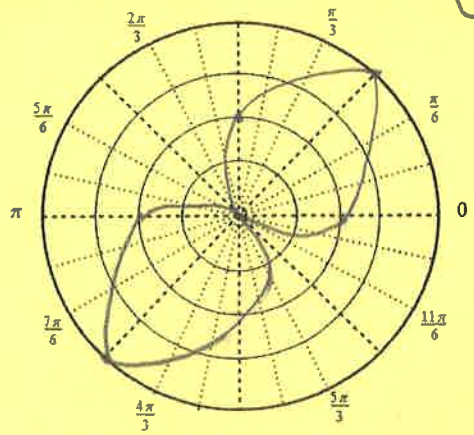
$r^2 = x^2 + y^2$
$= 1 + 3 = 4$
$r = -2$

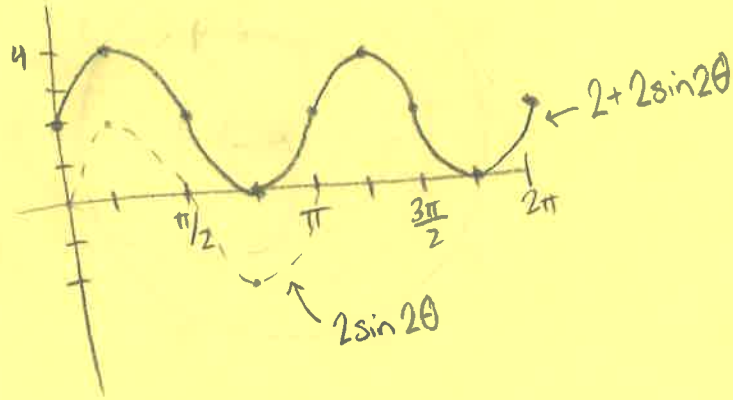
$\theta = \arctan\left(\frac{y}{x}\right)$
$= \arctan\left(\frac{\sqrt{3}}{-1}\right)$
$= \arctan(-\sqrt{3})$
$= \cancel{\frac{2\pi}{3}}, \frac{5\pi}{3}$



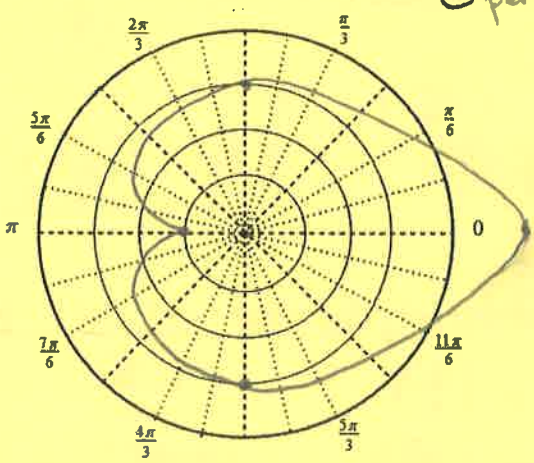
$\boxed{\left(-2, \frac{5\pi}{3}\right)}$

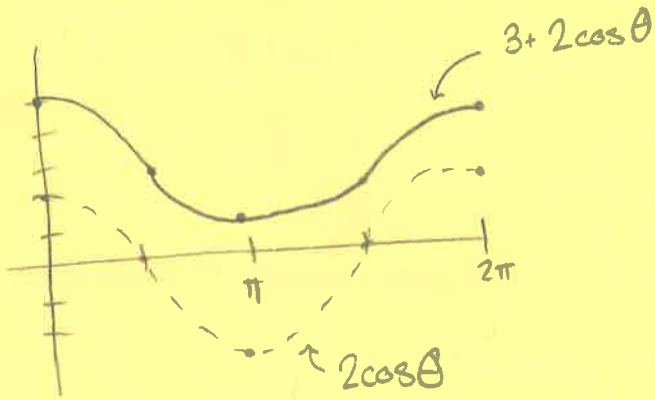
6. Sketch the graph. $r = 2 + 2\sin 2\theta$. period $= \frac{2\pi}{2} = \pi$



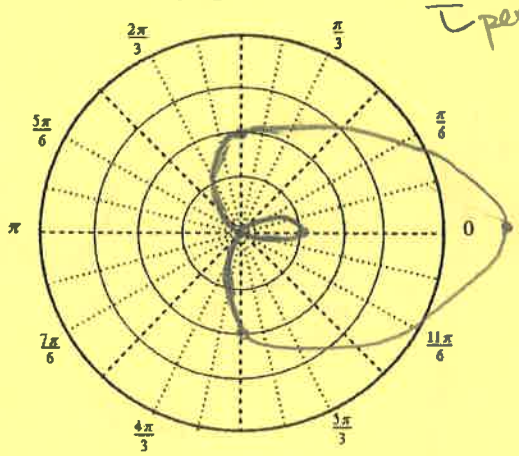


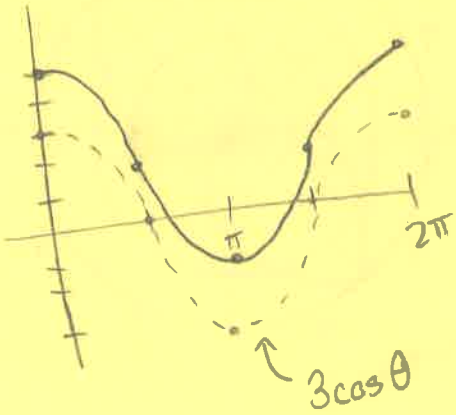
7. Sketch the graph. $r = 3 + 2\cos\theta$ period $= 2\pi$





8. Sketch the graph. $r = 2 + 3\cos\theta$ period $= 2\pi$





MATH 152 **QUIZ11** NAME Solutions

Mrs. Bonny Tighe 25 points

10.1,10.2,10.3 TA_____ Wed 12/6/2017

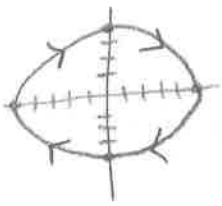
1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 5\sin\theta,\ \ y = 4\cos\theta$

$\frac{x}{5} = \sin\theta,\ \frac{y}{4} = \cos\theta$

$\left(\frac{x}{5}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$

$\frac{x^2}{25} + \frac{y^2}{16} = 1$



To figure out direction:

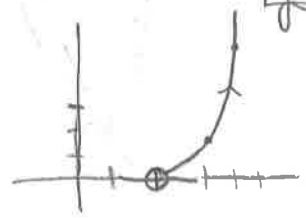
θ	x	y
0	0	4
$\pi/2$	5	0

$(0,4) \to (5,0)$

b) $x = 2 + e^t,\ \ y = e^{2t} = (e^t)^2$

$e^t = x - 2$

$\boxed{y = (x-2)^2}$



To figure out direction:

t	x	y
0	3	1
1	$2+e$	e^2

$2+e > 5$, $e^2 \approx 7.3$

Observe $y = e^{2t} \neq 0$ (hence the hole)

and $x = 2 + e^t > 2$ (hence no left half of parabola)

2. Find the equation of the tangent line to the curve $x = t^2,\ \ y = t^3$ at $(1, -1)$

$\frac{dx}{dt} = 2t \quad \frac{dy}{dt} = 3t^2$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2}{2t} = \frac{3}{2}t$

$m = \left.\frac{dy}{dx}\right|_{t=-1} = -\frac{3}{2}$

What is t at $(x,y) = (1,-1)$?

$x = t^2 = 1 \quad y = t^3 = -1$

$t = \cancel{x}, -1 \quad t = -1$

$y - y_1 = m(x - x_1)$

$y + 1 = -\frac{3}{2}(x - 1)$

$\boxed{y = -\frac{3}{2}x + \frac{1}{2}}$

3. Find the area enclosed by the x- axis and the curve $x = t^3 + 1,\ y = 2t - t^2$

$\int y\,dx$

$x = t^3 + 1$

$dx = 3t^2\,dt$

limits of integration: when curve crosses the x-axis

$y = 0 \Rightarrow 0 = 2t - t^2$

$= t(2-t)$

$t = 0, 2$

$$\int y\,dx = \int_0^2 (2t - t^2)(3t^2\,dt)$$

$$= \int_0^2 (6t^3 - 3t^4)\,dt$$

$$= \left(\frac{3}{2}t^4 - \frac{3}{5}t^5\right|_0^2$$

$$= \frac{3}{2}(2)^4 - \frac{3}{5}(2)^5 = 3(2)^3\left(1 - \frac{4}{5}\right) = 24\left(\frac{1}{5}\right) = \boxed{\frac{24}{5}}$$

4. Compute the surface area obtained by rotating the curve

$x = e^t + e^{-t},\ y = 5 - 2t,\ 0 \le t \le 3$ about the y-axis. $S = 2\pi \int_\alpha^\beta radius(arc\ length)$

$r = x = e^t + e^{-t}$

$\frac{dx}{dt} = e^t - e^{-t}$ $\frac{dy}{dt} = -2$

$$S = 2\pi \int_0^3 (e^t + e^{-t})\sqrt{(e^t - e^{-t})^2 + (-2)^2}\,dt$$

$$= 2\pi \int_0^3 (e^t + e^{-t})\sqrt{e^{2t} - 2 + e^{-2t} + 4}\,dt$$

"one of those"

$$= 2\pi \int_0^3 (e^t + e^{-t})\sqrt{(e^t + e^{-t})^2}\,dt$$

$$= 2\pi \int_0^3 (e^t + e^{-t})^2\,dt$$

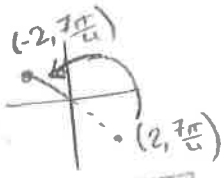
$$= 2\pi \int_0^3 (e^{2t} + 2 + e^{-2t})\,dt$$

$$= 2\pi \left(\frac{1}{2}e^{2t} + 2t - \frac{1}{2}e^{-2t}\right|_0^3$$

$$= 2\pi \left(\frac{1}{2}e^6 + 6 - \frac{1}{2}e^{-6} - \frac{1}{2} - 0 + \frac{1}{2}\right)$$

$$= \boxed{2\pi \left(\frac{1}{2}e^6 + 6 - \frac{1}{2}e^{-6}\right)}$$

5. a) Find the Cartesian coordinates for the polar coordinate $(-2, 7\pi/4)$



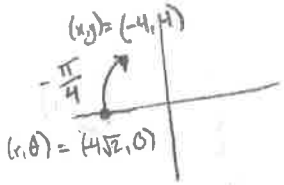
$x = r\cos\theta$
$= -2\cos\left(\frac{7\pi}{4}\right)$
$= -2\left(\frac{\sqrt{2}}{2}\right)$
$= -\sqrt{2}$

$y = r\sin\theta$
$= -2\sin\left(\frac{7\pi}{4}\right)$
$= -2\left(-\frac{\sqrt{2}}{2}\right)$
$= \sqrt{2}$

$\boxed{(x,y) = (-\sqrt{2}, \sqrt{2})}$

b) Find polar coordinates for the Cartesian coordinate (- 4, 4) where $\underline{r < 0, and\ \theta < 0}$

$r^2 = x^2 + y^2$
$= 16 + 16 = 32$
$r = -4\sqrt{2}$

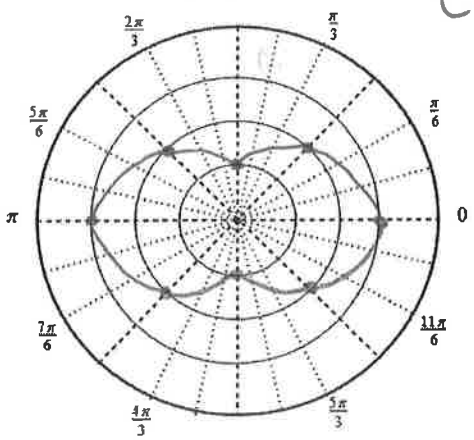


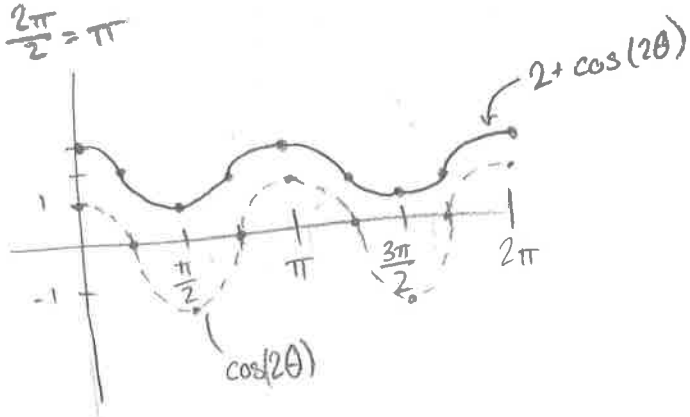
$\theta = \arctan\left(\frac{y}{x}\right)$
$= \arctan\left(\frac{4}{-4}\right)$
$= \arctan(-1)$
$= \frac{3\pi}{4}, \cancel{\frac{7\pi}{4}}$

$\boxed{\left(-4\sqrt{2}, -\frac{\pi}{4}\right)}$

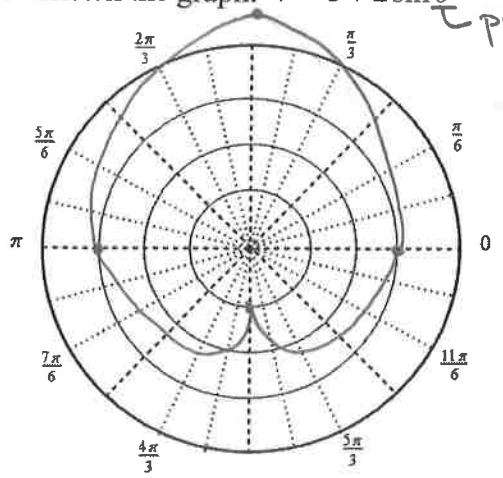
6. Sketch the graph. $r = 2 + \cos 2\theta$.

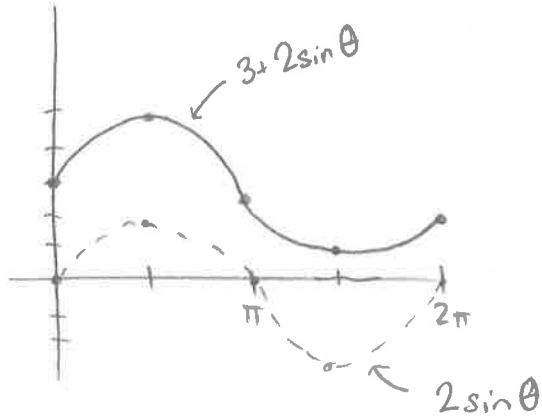
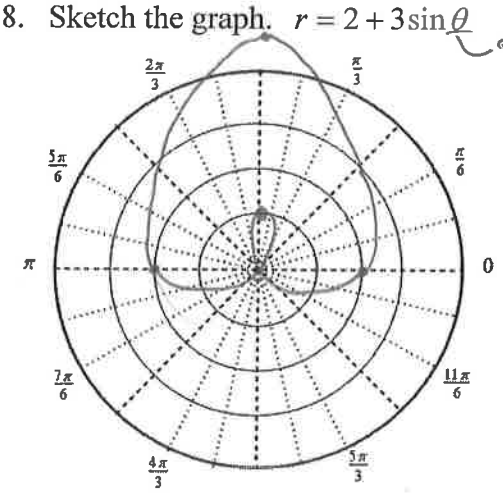
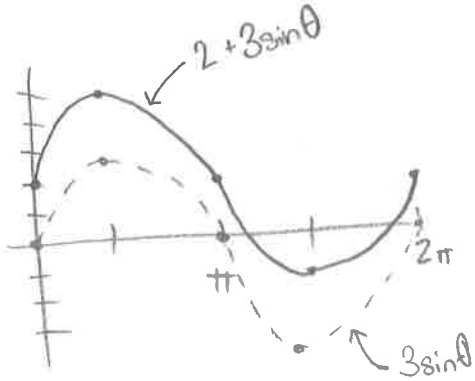
period $= \frac{2\pi}{2} = \pi$





7. Sketch the graph. $r = 3 + 2\sin\theta$ period $= 2\pi$

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